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Literature Review

Ethical Aspects of Synthetic Data in Smart Meter Datasets

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Abstract

This literature review examines the ethical dimensions of synthetic data generation in the context of household smart meter datasets and Non Intrusive Load Monitoring (NILM) research. Smart meter data occupies a peculiar position: technically unremarkable in its raw form which is a sequence of power consumption readings and yet capable of revealing occupancy patterns, daily routines, appliance inventories, and behavioural signatures that most people would consider deeply private. Synthetic data generation has been proposed as a route to enabling open research while protecting this privacy, but the literature reviewed here suggests that this promise is conditional and frequently overstated. Drawing on over thirty publications spanning energy informatics, machine learning privacy research, data ethics, and EU regulatory guidance, this review identifies four interconnected themes that will guide the research project: (1) the landscape of synthetic data generation methods for energy time series and what each approach achieves in terms of statistical fidelity; (2) the nature and empirical measurability of privacy risks that persist in synthetic energy data, including re-identification and attribute disclosure; (3) the under examined question of demographic and socioeconomic bias in smart meter datasets and how synthetic generation affects representational equity; and (4) the regulatory and ethical framework within which synthetic energy data must be evaluated, particularly under GDPR. The review concludes by formulating a research roadmap with six specific goals that build on identified gaps in the literature and reflect both the technical and ethical dimensions of the problem.

Keywords: synthetic data, smart meter datasets, NILM, privacy, GDPR, bias, energy disaggregation, generative models, data ethics, re-identification

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1. Introduction

Firstly, before beginning this literature review, it is worth acknowledging a tension that runs through all the sources I have read in preparing it. Smart meter data is, on the surface, a technical measurement in volts, amps, kilowatt-hours logged at regular intervals. It is the kind of data that engineers work with every day, and the kind that researchers need in order to develop better energy management systems, more accurate disaggregation algorithms, and smarter demand response mechanisms. Yet from the perspective of the person living in the metered household, this same data is a fine grained record of domestic life: when they shower, whether they are home, which appliances they own and use, and what their daily routine looks like. Recognising this duality which is technical measurement on one side and personal biography on the other is, in my view, the starting point for any honest engagement with the ethics of smart meter research.

Synthetic data has emerged as a proposed solution to the problem of sharing energy data for research without exposing individual households to privacy harm. Rather than releasing real consumption recordings, researchers and dataset providers can instead release algorithmically generated data that resembles the real data statistically but was not drawn from any actual household. This is an appealing idea, and there is genuine evidence that it works in important respects like the synthetic NILM dataset published by Klemenjak et al. (2020), for example, has been used in published research and has enabled work that would have required sensitive real data otherwise. But the literature also contains a growing body of evidence that synthetic data is not the clean privacy solution it is sometimes presented as, and that the ethical challenges do not disappear when data is generated rather than collected. They shift shape, and in some cases they intensify.

This literature review is organised around the four research objectives set out for this project. After establishing the landscape of smart meter datasets and the NILM context in Section 2, the review examines synthetic data generation methods in Section 3, focusing on what each approach achieves in terms of fidelity, privacy, and representational equity. Section 4 addresses the privacy risks that persist in synthetic energy data and the empirical methods available to assess them. Section 5 turns to the under examined question of demographic bias in energy datasets and how synthetic generation affects it. Section 6 consolidates the research gaps identified across these themes, and Section 7 translates these gaps into a concrete research roadmap with six specific, achievable goals.

A note on sources: this review draws primarily on peer reviewed publications in energy informatics, machine learning, and privacy engineering, supplemented by regulatory guidance from the Article 29 Working Party and the GDPR itself. Where possible, I have prioritised sources that engage empirically with the specific challenges of energy time series data rather than making general claims about synthetic data that may not hold in this domain. The two anchor datasets which are SmartNIALMeter (Vogel et al., 2024) and the Klemenjak et al. (2020) synthetic dataset are treated as concrete reference points throughout rather than as abstract examples.

2. Smart Meter Datasets: Landscape, Characteristics, and Privacy Stakes

2.1 The Smart Meter Data Landscape in Europe

Smart electricity meters are devices capable of recording energy consumption at high temporal resolution typically every 15 or 30 minutes and transmitting this data to utilities and system operators. Their mass deployment across Europe has been driven by the European Commission's smart metering directives, which set targets for rollout across member states as part of broader smart grid infrastructure programmes (European Commission, 2012). By the mid 2020s, tens of millions of households across the EU were equipped with smart meters, making them one of the largest continuous sensing infrastructures in civilian use.

From a data perspective, what smart meters produce is a time series: a sequence of power consumption values indexed by timestamp, typically at sub hourly resolution. For research purposes, this creates an enormous and inherently longitudinal dataset describing real patterns of domestic energy use. The scientific value is substantial: researchers can study seasonal demand patterns, evaluate appliance efficiency, train disaggregation models, and develop demand response systems in ways that were previously only possible through expensive bespoke studies. Publicly available smart meter datasets have become important resources in this research community, and their quality and coverage have improved substantially over the past decade.

However, the same features that make smart meter data scientifically valuable also make it privacy sensitive. The temporal resolution and longitudinal continuity of smart meter data means it encodes behavioural information at a granularity that is difficult to de identify or remove. Unlike a one time medical measurement, smart meter data accumulates over months and years, creating a detailed behavioural signature for each household. Beckel et al. (2014) demonstrated that simple statistical features extracted from smart meter time series like mean consumption, consumption variance, peak time distributions are sufficient to predict household size, occupancy patterns, and even socioeconomic indicators with meaningful accuracy. Molina Markham et al. (2010) showed that inference attacks exploiting the temporal structure of aggregate consumption signals could recover information about individual appliance usage even without access to sub metered data. The

conclusion drawn from these and subsequent studies is clear: smart meter data is personal data in the substantive sense, regardless of whether it is linked to a named individual.

2.2 Non Intrusive Load Monitoring and the Privacy Implication

Non Intrusive Load Monitoring (NILM) is the computational task of disaggregating a household's aggregate electricity consumption signal into appliance level usage estimates. The term 'non intrusive' refers to the fact that NILM requires only a single measurement point at the main electricity meter, without requiring separate sensors on individual appliances. The task is an inverse problem: given the aggregate signal, recover the individual contributions of each appliance.

NILM was formalised as a research problem by Hart (1992), who proposed modelling each appliance as a finite state machine with distinct power consumption levels and using transitions in the aggregate signal to identify appliance state changes. This combinatorial optimisation approach formed the basis of NILM research for nearly two decades. Contemporary approaches have largely shifted towards probabilistic and machine learning methods. Kolter and Jaakkola (2012) applied additive factorial Hidden Markov Models to the disaggregation problem, treating each appliance as a latent Markov chain and using variational inference to recover appliance states from the aggregate signal. Kelly and Knottenbelt (2015) proposed Neural NILM, demonstrating that sequence to point convolutional networks could outperform HMM based methods on the UK DALE benchmark dataset. More recent work has applied transformer based architectures and self supervised pretraining to the disaggregation problem, continuing to push accuracy metrics upward (Nalmpantis and Vrakas, 2019).

The relevance of NILM to the ethics of smart meter data is direct and important. Every improvement in NILM performance increases the amount of behavioural information that can be extracted from aggregate consumption data. A household's aggregate smart meter reading reveals that they consumed 3.2 kWh between 6 pm and 8 pm; a NILM algorithm applied to that reading can in principle reveal that 1.1 kWh went to the electric oven, 0.8 kWh to the dishwasher, and 0.4 kWh to a television. The granularity of inference that NILM enables means that privacy risk is not bounded by the resolution of the smart meter itself but by the sophistication of the algorithms that

can be applied to it. This has significant implications for the evaluation of synthetic data: a synthetic dataset that appears to protect privacy at the aggregate level may still preserve the appliance level temporal signatures that NILM exploits, enabling the extraction of sensitive behavioural information from synthetic data.

2.3 Anchor Datasets: SmartNIALMeter and Klemenjak et al.

Two datasets serve as the primary empirical anchors for this research project, and it is worth examining them carefully here because much of what follows depends on understanding what they contain and how they were produced.

SmartNIALMeter (Vogel et al., 2024): The SmartNIALMeter dataset is a real household appliance disaggregation dataset described by Vogel et al. (2024) in Data in Brief. The dataset records aggregate household electricity consumption alongside individual appliance sub metering at high temporal resolution from consenting households. As a dataset of genuine residential measurements, SmartNIALMeter carries the full privacy weight described in Section 2.1: it contains real behavioural signatures of real people, captured with their consent under a documented data management protocol. What makes the dataset particularly valuable for this research is that it provides a well documented real world reference point against which synthetic datasets can be systematically compared as this is a role that is surprisingly difficult to fill well in the NILM literature, where many real datasets are either not publicly available or are available only under conditions that prevent direct empirical evaluation.

Klemenjak et al. Synthetic Dataset (2020): The synthetic dataset for NILM published by Klemenjak, Kovatsch, Herold, and Elmenreich (2020) in Scientific Data represents a different approach. Rather than collecting data from households, the authors constructed explicit appliance models from statistics derived from measurement campaigns and published literature, and generated synthetic consumption signals by simulating these models. The dataset provides both aggregate signals and sub metered appliance signals for a set of synthetic households, enabling it to serve as a training and evaluation resource for NILM research. Crucially, the paper includes extensive validation comparing the synthetic signals to real world equivalents on multiple statistical and task performance measures, making it one of the few synthetic energy datasets with

documented empirical fidelity evaluation. The fact that this dataset was co authored by Prof. Elmenreich, the supervisor of this research project, adds a direct connection between the research context and one of its primary sources.

Together, these two datasets provide the empirical foundation for the research. The SmartNIALMeter data can serve as a source of ground truth real measurements, and the Klemenjak et al. synthetic data provides a baseline synthetic comparison. The research will use both to evaluate the claims made in the literature about the privacy and fidelity properties of synthetic energy data.

3. Synthetic Data Generation: Methods and Trade offs

One of the first things that becomes clear when surveying the literature on synthetic energy data is that 'synthetic data' is not a single method but a family of approaches with substantially different properties. The differences matter for ethics: different generation methods expose different residual privacy risks, produce different kinds of bias, and offer different degrees of controllability. A thorough understanding of the methodological landscape is a prerequisite for any rigorous ethical evaluation.

3.1 Statistical and Model Based Approaches

The oldest and most transparent class of synthetic energy data generation methods uses explicit statistical models of appliance behaviour. The core idea is to characterise each appliance type as a probabilistic state machine: a device that transitions between operating states (off, standby, active, varying) with specified transition probabilities, and that draws power from a specified distribution in each state. Synthetic household consumption data is then generated by simulating multiple appliances independently and combining their contributions into an aggregate signal.

Klemenjak et al. (2020) implement this approach using a combination of published appliance characterisations and data from laboratory measurements. Their generator constructs a library of appliance models each defined by state transitions, power distributions, and usage schedule priors and simulates these to produce synthetic sub metered and aggregate signals. The resulting dataset has been validated against real data on statistical measures including mean consumption distributions, load profile shapes, and NILM algorithm performance, with generally favourable results.

Richardson et al. (2010) developed an influential model along similar lines specifically for residential demand simulation, using time use survey data to drive occupancy and activity schedules that in turn determine appliance usage. This occupancy driven approach generates synthetic data that is anchored to observed human behaviour rather than purely electrical characteristics, producing patterns with realistic circadian and weekly structure.

The methodological advantages of model based approaches are interpretability and controllability. The researcher knows exactly what assumptions are encoded in the model and can adjust parameters to explore different household configurations. The limitations are equally clear: the model can only represent what its architecture is capable of expressing. Complex, context dependent appliance behaviours and the way a household's cooking pattern on weeknights differs from weekends in ways that are difficult to reduce to transition probabilities may be absent from the synthetic output even when they are present and consistent in real data. Additionally, the model parameters must be estimated from real data somewhere, creating a potential if indirect pathway through which real household information can influence the synthetic output.

3.2 Generative Deep Learning: GANs and VAEs

The past decade has seen the widespread application of deep generative models particularly Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) to the problem of synthetic data generation. These methods learn a generative model implicitly from training data, without requiring the researcher to specify explicit statistical assumptions. For time series data like energy consumption, architectures specifically designed to preserve temporal structure have been developed, most notably TimeGAN (Yoon et al., 2019), which combines an adversarial training objective with a supervised objective on temporal dynamics.

In the energy domain, GAN based methods have been applied to residential load profiles at the aggregate household level. Relevant work has demonstrated that well trained energy domain GANs can reproduce marginal consumption distributions and diurnal patterns with high fidelity, and that the generated profiles are difficult to distinguish from real profiles on standard statistical tests. Wang et al. (2019) proposed a conditional GAN framework for generating realistic residential load profiles conditioned on household characteristics, demonstrating that the generation process could be controlled to produce profiles for specified household types. Kempitiya et al. (2020) applied similar approaches to the sub metered appliance level, attempting to generate synthetic appliance signatures suitable for NILM training.

The practical challenges associated with GAN based generation are well documented in the literature. GAN training is notoriously unstable: training dynamics are sensitive to hyperparameter

choices, architecture decisions, and the scale and diversity of training data. Mode collapse which is the tendency of GANs to generate samples concentrated near the modes of the training distribution rather than covering its full diversity is a persistent problem with direct implications for bias: a GAN trained on a dataset that includes ten different household types may effectively generate only the most common two or three, discarding the diversity of the training data in its output. The privacy properties of GAN generated data are also difficult to characterise analytically, a point addressed in detail in Section 4.

VAEs offer a different set of trade offs. By learning a structured latent space that represents the generative factors of variation in the training data, VAEs produce smoother and more diverse outputs than GANs and are more stable to train. The probabilistic encoder decoder architecture also provides a more principled framework for controlling generation. However, VAE outputs often exhibit blurring of fine temporal structure like the sharp peaks and transitions that characterise appliance switching events which is precisely the information that NILM algorithms need to function, making VAEs less well suited to generating NILM training data than GAN based approaches.

3.3 Agent Based and Occupancy Driven Models

A third family of methods generates synthetic energy data by simulating human behaviour rather than directly modelling electrical signals. Agent-based models assign simulated occupants to households, give them activity schedules derived from time-use survey data or behavioural models, and derive appliance usage and energy consumption from these activity patterns. This approach produces synthetic data with a realistic social and behavioural structure, because the consumption patterns emerge from simulated human activities rather than from direct statistical modelling of electrical signals.

The Richardson et al. (2010) model mentioned in Section 3.1 is a foundational example of this approach. More recent work has explored more sophisticated agent based frameworks that incorporate household composition, social norms, and seasonal variation, producing synthetic datasets with richer variation than purely electrical models can achieve. Fischer (2008) developed an early comprehensive model along these lines, establishing the template that subsequent work

has refined. The approach is particularly relevant to the bias analysis in Section 5, because the time use survey data that drives activity schedules in these models reflects the demographic composition of survey respondents thats if that population is not representative of the general population, the synthetic data generated from it will inherit those representational gaps.

3.4 Comparative Evaluation: What Each Method Offers

Drawing the three approaches together, a clear comparative picture emerges. Table 1 summarises the key properties of each class of generation method along the three dimensions most relevant to this research: statistical fidelity (how closely the synthetic data resembles real data), privacy properties (how well the generation process protects against re-identification and attribute disclosure), and fairness risk (how likely the method is to inherit or amplify demographic biases from source data).

Method	Key Examples	Statistical Fidelity	Privacy Properties	Bias / Fairness Risk
Statistical / Model Based	Klemenjak et al. (2020); Richardson et al. (2010)	Good for modelled patterns; misses unusual behaviours	Low if parameters from aggregates; risk if trained on individual records	Inherits bias of model parameterisation source data
GAN / Deep Generative	TimeGAN (Yoon et al., 2019); Wang et al. (2019)	High for common patterns; mode collapse risk	No formal guarantee without DP; susceptible to membership inference	Mode collapse may amplify majority-pattern over-representation
VAE	Kingma & Welling (2014) adapted for energy	Smooth but misses sharp appliance transients	Similar to GAN; no formal guarantee	More diverse output; lower mode collapse risk
Agent Based / Occupancy Driven	Fischer (2008); Richardson et al. (2010)	Realistic behavioural structure; diversity by design	Strong if survey data is aggregate; weak if individual-level	Upstream bias from time-use survey demographics
Differential Privacy (any)	Combined with GAN, VAE, or statistical	Degrades with stronger ε; temporal structure most affected	Formal guarantee via ε; gold standard	DP noise does not address

				representational bias

Table 1: Comparative overview of synthetic energy data generation methods across fidelity, privacy, and fairness dimensions.

Several things stand out from this comparison. First, no method dominates across all three dimensions. Model based methods have the best interpretability and controllability but may miss important patterns; deep generative models can capture complex distributions but offer no privacy guarantees; differential privacy provides formal guarantees but at the cost of temporal fidelity. Any research evaluation of synthetic energy data must therefore be explicit about which dimensions it is optimising and which trade offs it is accepting.

Second, the privacy properties of GAN based methods are particularly concerning and are addressed at length in Section 4. The key point here is that statistical similarity to real data which is what most fidelity metrics measure does not imply privacy protection. A GAN that has nearly memorised its training data will score well on fidelity metrics while offering effectively no privacy. This dissociation between fidelity and privacy is one of the central problems in the ethics of synthetic data.

4. Privacy Risks in Synthetic Energy Data

The strongest version of the case for synthetic data is that it enables research to happen without privacy risk: the data was never collected from real people, or was transformed so thoroughly that no real person's information can be recovered from it. This section examines the evidence for and against this claim in the specific context of smart meter data, and concludes that while synthetic data meaningfully reduces some privacy risks, it does not eliminate them, and that treating it as inherently private is an assumption the literature does not support.

4.1 The Myth of Automatic Anonymisation

Perhaps the most important single paper for situating the privacy debate around synthetic data is Stadler, Oprisanu, and Troncoso (2022), 'Synthetic Data, Anonymisation Groundhog Day.' The paper systematically evaluates the privacy properties of synthetic datasets generated by a range of standard methods including GANs and statistical models and arrives at a striking conclusion: standard synthetic data generators trained without differential privacy provide no formal privacy guarantees, and are vulnerable to the same categories of privacy attack as pseudonymised real data. The paper demonstrates that membership inference attacks that determine whether a specific individual's data was present in a generator's training set can succeed against synthetic data generators with non trivial accuracy well above random chance, even when the generator has not obviously memorised specific records.

This finding is directly applicable to the energy domain. If a GAN is trained on smart meter data from a specific set of households, an adversary with knowledge of one of those households' consumption patterns can, in principle, determine whether that household was in the training set by probing the generator's outputs. This is a violation of privacy even if the synthetic outputs themselves do not directly reveal consumption values which is a fact of participation in a data collection exercise may itself be sensitive.

The mechanisms behind this vulnerability are subtle but well understood. Machine learning models, including generative models, tend to encode properties of their training data in their weights. This is, in fact, what makes them useful: a GAN trained on residential load profiles learns

what residential load profiles look like, and this learned knowledge is encoded in its parameters. But the same mechanism means that, if the training data included a household with an unusual consumption pattern like a night-shift worker, a household with an electric vehicle, a home with medical equipment, then that pattern may be reproduced with unusual fidelity in the generator's output, creating a pathway through which specific individuals' patterns can be recovered.

4.2 Privacy Attack Taxonomy

It is useful to distinguish between the main categories of privacy attack that are relevant to synthetic energy data, because they operate through different mechanisms and have different implications for evaluation methodology.

Membership inference attacks ask whether a specific individual's data was present in the training set used to build a synthetic generator. In the energy context, a successful membership inference attack would reveal that a specific household participated in the data collection exercise underlying a synthetic dataset. Shokri et al. (2017) demonstrated general methods for membership inference against machine learning models that have since been applied to generative models, and their shadow modelling technique like training auxiliary models on datasets with and without the target record has become a standard evaluation methodology. For model based generators like Klemenjak et al. (2020), standard shadow modelling does not directly apply because the generator is not a neural network trained end to end; however, analogous sensitivity analyses over generator parameters can reveal whether specific individuals' data influenced the generation.

Attribute disclosure attacks attempt to infer sensitive properties of individuals from the synthetic data, without necessarily identifying those individuals. In the energy context, relevant sensitive attributes include: occupancy during working hours (indicating home working or unemployment), nocturnal consumption patterns (inferring sleep schedule), specific appliance signatures (revealing medical equipment, grow lamps, or electric vehicle charging), and weekend versus weekday consumption differences (inferring working schedule). Even when a synthetic dataset does not contain direct pointers to real individuals, it may preserve the statistical regularities from which these attributes can be inferred like a synthetic household that has the characteristic signature of

an elderly person living alone may enable attribute inference even if no real elderly person's data is directly encoded.

Re identification attacks attempt to match synthetic records to real individuals by comparing synthetic outputs to external datasets or auxiliary information. In the smart meter context, an adversary with access to publicly available aggregate energy statistics for a neighbourhood, or to another dataset that can be spatially joined to the synthetic data, might be able to narrow down which real household a synthetic record is most similar to. The risk is lower for synthetic data than for pseudonymised real data, because synthetic records are not transformed versions of real records. However, if the generator systematically reproduces the statistical properties of specific households as can occur with mode collapse or when training sets are small then the risk becomes non negligible.

4.3 Empirical Privacy Metrics

A significant practical challenge in evaluating the privacy of synthetic energy datasets is the absence of standardised metrics. The literature contains several proposals, but there is no established community standard analogous to, say, RMSE for prediction accuracy or F score for classification. This gap motivates one of the concrete research goals articulated in Section 7.

The Nearest Neighbour Distance Ratio (NNDR) is perhaps the most widely used empirical privacy metric for synthetic datasets. It measures, for each synthetic record, the ratio of the distance to its nearest real record to the distance to its nearest synthetic neighbour. Low NNDR values indicate synthetic records that are very similar to specific real records, flagging potential memorisation by the generator. The metric has been applied in the tabular data context and can be adapted to energy time series data with appropriate distance functions.

Membership inference advantage is the accuracy advantage of a membership inference classifier over random chance which provides a more operationally grounded privacy measure. It directly asks: how much can an adversary learn about training set membership from access to the synthetic data? An advantage near zero indicates strong privacy; a larger advantage indicates that the generator has encoded training set membership information in its outputs. Computing this metric

for energy domain generators is methodologically straightforward for neural network generators but requires adaptation for model based generators.

Attribute inference accuracy, measured against population level base rates, quantifies the degree to which sensitive household attributes can be inferred from synthetic records. For energy data, candidate attributes include occupancy patterns, household size indicators, and appliance inventory features. A synthetic dataset for which attribute inference accuracy substantially exceeds the population base rate has, in effect, preserved the inference channels through which real households' attributes can be extracted.

4.4 GDPR Compliance and the Anonymisation Threshold

The European Union's General Data Protection Regulation (GDPR), enacted in 2018, establishes the legal framework within which smart meter data must be handled in EU member states (European Parliament and Council, 2016). Smart meter data is unambiguously personal data under GDPR Article 4(1): it relates to an identified or identifiable natural person, namely the resident of the metered address.

The key question for synthetic data is whether synthetic generation transforms the data sufficiently to render it no longer personal data and therefore no longer subject to GDPR's requirements. Recital 26 of the Regulation addresses this: data that has been 'rendered anonymous in such a manner that the data subject is not or no longer identifiable' falls outside the scope of the Regulation. This is the provision that proponents of synthetic data often invoke: they argue that generating synthetic data constitutes anonymisation, removing GDPR applicability.

The Article 29 Working Party's Opinion 05/2014 on anonymisation techniques (WP216) provides important regulatory guidance on the conditions under which this argument holds. WP216 identifies three tests for genuine anonymisation: the data should prevent singling out (uniquely identifying an individual), linkability (matching records from different datasets), and inference (deriving new information about an individual). Critically, WP216 emphasises that these tests must be applied considering realistic, not hypothetical, adversaries that an adversary who has access to auxiliary information that a reasonable person could foresee being available. For energy data, this

includes the possibility of spatial location information, utility records, and published aggregate statistics.

The implication for synthetic data is that claiming GDPR compliance on the basis of synthetic generation requires demonstrating, through empirical evaluation, that the resulting dataset satisfies all three WP216 tests under realistic adversarial assumptions. Simply running a GAN and asserting that the output is synthetic and therefore anonymous is not sufficient. This is a point that several recent papers have made forcefully (Stadler et al., 2022; Rankin et al., 2020), and it motivates the empirical evaluation methodology proposed in the research roadmap in Section 7.

5. Bias and Representational Fairness in Energy Datasets

If the privacy literature on synthetic data is relatively well developed like the Stadler et al. (2022) paper has been widely cited and has substantially changed the field's understanding of what synthetic data achieves then the question of demographic bias and representational fairness in synthetic energy data has received far less systematic attention. This is a genuine gap in the literature, and one that this research project is well positioned to address. The following three subsections examine the sources of bias in real smart meter datasets, the mechanisms through which synthetic generation can affect these biases, and the implications for energy justice.

5.1 Sources of Bias in Real Smart Meter Datasets

Real smart meter datasets, including those used to train or parameterise synthetic generators, are not representative samples of the population. They are convenience samples, shaped by the practical realities of data collection. Understanding the specific sources of bias in these datasets is a prerequisite for assessing how synthetic generation will propagate or modify them.

Collection bias: Most publicly available smart meter datasets were collected through voluntary participation programmes like the research projects, utility pilot schemes, or opt in monitoring initiatives. Voluntary participation in energy monitoring research is systematically skewed. It is disproportionately common among owner occupiers (renters often lack the standing or motivation to participate in monitoring programmes that primarily benefit property owners), among higher income households (who have stronger financial incentive to monitor and optimise their energy use), and among technology literate demographics. The SGSC dataset used by Kong et al. (2019) in the NILM context, the UK DALE dataset used by Kelly and Knottenbelt (2015), and many other reference datasets in the field were collected through exactly these mechanisms.

Appliance inventory bias: Smart meter datasets for NILM research tend to be collected from households that have the specific appliances targeted by the disaggregation algorithms under development. This creates a feedback loop: researchers collect data from households with dishwashers, washing machines, and electric ovens, develop algorithms for those appliances, and validate on similar datasets and never engaging with the very different load profiles of households

that have different appliance inventories, different cooking technologies (induction vs. gas), or different heating systems. The Klemenjak et al. (2020) synthetic dataset was parameterised from real measurement campaigns that were themselves subject to these biases, and it is reasonable to expect that the synthetic data inherits the appliance inventory of the source campaigns.

Geographic and climatic bias: Energy consumption patterns are strongly shaped by climate, housing stock, and infrastructure. A dataset collected in the Austrian Alps will reflect the specific climatic profile, building construction norms, and heating technology mix of that region. Synthetic data generated from such a dataset will produce outputs that resemble that regional profile rather than the general population of European or global households. This matters when synthetic data is used to train NILM algorithms or evaluate demand response systems that will be deployed across diverse contexts.

Temporal bias: Many benchmark datasets cover relatively short periods like a few months to a year and may not represent the full seasonal and annual variability of household consumption. Synthetic data trained on such datasets may fail to capture the long term patterns and inter annual variability that are important for some research questions.

5.2 How Synthetic Generation Affects Bias

The question of how synthetic generation affects the biases described in Section 5.1 does not have a single answer: the effect depends heavily on the generation method used and the choices made in designing and training the generator.

For model based and statistical generation methods, the relationship between source bias and output bias is relatively transparent. If the appliance models are parameterised from measurement data that over represents high income households with large appliance inventories, the synthetic dataset will produce households with similar inventories. The bias is inherited proportionally, but it is at least identifiable: a careful analyst can trace the origin of the synthetic data's characteristics back to the parameterisation decisions. This is the approach used by Klemenjak et al. (2020), and it is one of the reasons that their dataset documentation is relatively informative about its representational scope.

For deep generative models, particularly GANs, the relationship between source bias and output bias is more treacherous. Mode collapse which is the tendency of GAN generators to concentrate their output on the most common patterns in the training data which can amplify existing biases rather than merely reproducing them. If the training data contains 70% of households with a particular consumption profile and 15% each of two other profiles, a mode collapsed GAN might produce outputs that are 90% the majority profile and 5% each of the minority profiles, effectively erasing the diversity that was present in the original data. For energy research, this means that a GAN trained on a dataset that already under represents low income households with restricted energy use may produce a synthetic dataset in which those households are essentially absent.

The possibility of using synthetic generation to correct, rather than perpetuate, existing biases is one of the more interesting research questions in this area. Because the researcher controls the generation process, they can in principle configure it to produce a more demographically representative dataset than the source data allows deliberately generating synthetic households for demographic profiles that are under represented in the original data. This possibility is more naturally implemented in agent based and model based approaches, where household characteristics can be specified as parameters, than in data driven deep generative models where the output distribution is implicitly determined by the training data.

5.3 Energy Justice Implications

The bias concerns in the previous two sections are not merely methodological inconveniences. They have direct implications for what is sometimes called energy justice which is the equitable distribution of the benefits and burdens of energy systems across different social groups.

Research on energy consumption, energy efficiency, and appliance usage feeds into policy decisions: utility tariff structures, building efficiency standards, appliance labelling requirements, and demand response programme design. If the datasets underpinning this research systematically under represent low income households, renters, households in social housing, and households with older or less efficient appliances, then the research conclusions and the policies derived from them may be systematically miscalibrated for the population segments with the most acute energy challenges. This is not a hypothetical concern: there is documented evidence that energy poverty

research has been constrained by the availability of high quality data for lower income household segments (Deller et al., 2021; Klemenjak and Egarter, 2016).

The possibility that synthetic data could provide a route to correcting these representational gaps by deliberately generating diverse household profiles that include under represented demographic categories makes the fairness question practically important as well as ethically significant. This is an area where I believe the project can make a genuine contribution, and it is reflected in the research roadmap in Section 7.

6. Research Gaps and Open Questions

Some questions have been addressed rigorously and repeatedly in Sections 2 through 5 like the feasibility of NILM at various temporal resolutions, the architecture of GAN based time series generators, the statistical tests for measuring fidelity between real and synthetic distributions. Other questions have received relatively little systematic attention in the specific context of smart meter data. The following gaps, identified through reading, are directly relevant to the research project:

- **Gap 1: Absence of standardised privacy evaluation for energy synthetic data:** While privacy metrics such as NNDR and membership inference advantage have been proposed in the general synthetic data literature (Stadler et al., 2022; Jordon et al., 2022), they have not been systematically applied to energy time series data. The specific characteristics of smart meter data is its temporal autocorrelation, its appliance driven multimodality, its sensitivity to distance metric choice means that metrics developed for tabular data may not transfer directly. There is a clear need for a validated methodology for privacy evaluation specifically suited to energy consumption time series.

- **Gap 2: Inadequate validation of synthetic datasets for downstream NILM tasks:** Many synthetic energy datasets are validated against statistical properties of the training data (distribution similarity, autocorrelation structure) but not against the downstream NILM tasks for which they are intended. Fidelity on statistical metrics does not guarantee that NILM algorithms trained on synthetic data will generalise to real household signals, and the gap between these two forms of validation has not been systematically characterised in the literature.

- **Gap 3: Lack of demographic characterisation of synthetic energy datasets:** Very few published synthetic energy datasets include explicit documentation of the demographic and socioeconomic characteristics of the households they represent, or systematic comparison of their representativeness against census or survey data. Without this documentation, users of synthetic datasets have no way to assess whether the dataset is appropriate for their research context. This is both a data documentation gap and a research methods gap.

- **Gap 4: Limited exploration of bias correction in synthetic generation:** The possibility of using synthetic generation to deliberately correct demographic under representation in energy datasets has been discussed in general terms in the broader synthetic data literature (Jordon et al., 2022) but has not been explored empirically in the energy domain. It is not known whether current generation methods can produce demographically diverse synthetic datasets of sufficient quality, or what the fidelity trade offs of such an approach would be.

- **Gap 5: GDPR analysis of energy synthetic data generation pipelines:** While the general applicability of GDPR to synthetic data has been discussed (Rankin et al., 2020; Synthetic Data Institute, 2023), there is no published analysis specifically addressing the GDPR compliance of the energy synthetic data generation pipelines currently in use including the question of whether the collection of real household data for model parameterisation (as in Klemenjak et al., 2020) requires a specific legal basis or data subject notification.

- **Gap 6: Mode collapse characterisation for energy domain GANs:** GAN mode collapse and its effects on the diversity of generated energy profiles have not been systematically studied in the energy domain. It is not clear how frequently mode collapse occurs with current architectures, how severe its effects on representational diversity are, and whether standard collapse detection methods translate to the energy domain.

These six gaps directly motivate the research goals articulated in Section 7. Each goal is designed to address one or more of these gaps using the anchor datasets and analytical methods described in this review.

7. Research Goals and Roadmap

Having surveyed the relevant literature and identified the most significant open questions, I can now articulate the specific goals of this research project with greater precision than the initial project description allowed. The six goals below are ordered to reflect a natural analytical progression: understanding what synthetic data is and how it is generated before asking what risks it carries, and addressing individual risks before attempting to formulate integrated recommendations. Each goal is connected to a specific literature gap identified in Section 6.

Goal 1: Comparative Review of Synthetic Generation Methods

Research question: How do statistical, deep generative, and agent based synthetic data generation methods compare in their ability to reproduce realistic household energy consumption patterns, and what are the specific limitations of each approach for NILM applications?

Motivation: The literature review has established that different generation methods have substantially different properties, but head to head empirical comparisons in the specific context of NILM oriented energy data are rare. The Klemenjak et al. (2020) synthetic dataset uses a model based approach; the literature contains GAN based alternatives but no rigorous comparison against the same real data reference. This comparative analysis will establish the empirical baseline for the rest of the research.

Approach: Compare at least three generation methods including the Klemenjak et al. Model based approach, at least one GAN based method, and an occupancy driven approach against the SmartNIALMeter real dataset. Evaluation will use both statistical fidelity metrics (KS statistic, Earth Mover's Distance, autocorrelation function comparison) and downstream NILM performance (F score per appliance, normalised disaggregation error).

Links to gaps: Addresses Gap 2 (inadequate NILM task validation) and provides the empirical foundation for Goals 2–4.

Goal 2: Empirical Privacy Evaluation

Research question: What measurable privacy risks remain in the synthetic energy datasets produced by standard generation methods, and how do these risks compare to those present in the real data?

Motivation: The Stadler et al. (2022) finding that standard synthetic data generation provides no formal privacy guarantees has not been empirically replicated in the energy domain. Given the specific temporal structure of energy time series data, it is not clear whether the attack methods developed for tabular data transfer directly, and the relative effectiveness of different attack types against different generation methods has not been characterised.

Approach: Apply NNDR, membership inference advantage, and attribute inference accuracy metrics to both the Klemenjak et al. synthetic dataset and to at least one freshly generated synthetic dataset, comparing results to a baseline computed on the real SmartNIALMeter data. Sensitive attributes to target for inference will include occupancy pattern indicators and appliance inventory features.

Links to gaps: Directly addresses Gap 1 (absence of standardised privacy evaluation for energy synthetic data).

Goal 3: GDPR Compliance Assessment

Research question: Do the synthetic energy datasets produced by standard generation methods meet the anonymisation threshold required for GDPR compliance, as defined by the Article 29 Working Party Opinion WP216?

Motivation: Many researchers using synthetic smart meter data implicitly assume that synthetic generation confers GDPR compliance. The regulatory analysis in Section 4.4 suggests this assumption is not warranted without empirical evaluation. This goal aims to make the compliance assessment explicit and evidence based rather than assumed.

Approach: Apply the three WP216 tests (singling out, linkability, inference) to the synthetic datasets evaluated in Goal 2, documenting the assessment methodology and conclusions. This will involve both quantitative privacy metrics (from Goal 2) and qualitative assessment of realistic adversarial scenarios relevant to the energy context.

Links to gaps: Addresses Gap 5 (GDPR analysis of energy synthetic data pipelines).

Goal 4: Bias Analysis

Research question: What demographic and socioeconomic biases are present in the anchor datasets, and how are these biases affected by different synthetic generation approaches which means are they inherited, amplified, or potentially corrected?

Motivation: Section 5 identified demographic bias as an under examined dimension of synthetic energy data ethics. Characterising the biases present in the anchor datasets and quantifying how generation affects them will produce evidence-based findings that are largely absent from the current literature.

Approach: Profile the household characteristics represented in the SmartNIALMeter dataset and the Klemenjak et al. synthetic dataset against available census and demographic statistics for the relevant regions. Apply cluster analysis to the synthetic outputs to assess diversity and test for mode collapse effects. Evaluate whether deliberate adjustment of generation parameters can produce more demographically representative synthetic data.

Links to gaps: Addresses Gaps 3, 4, and 6.

Goal 5: Exploratory Data Analysis and Routine Identification

Research question: What daily routine patterns are identifiable in real and synthetic smart meter data, and what are the implications of these patterns for privacy assessment and utility evaluation?

Motivation: The project brief specifically identifies exploratory data analysis as an important starting task, and the routine identification question is central to the privacy evaluation: if routines can be isolated from both real and synthetic data, their presence in synthetic data constitutes a privacy relevant disclosure. This goal provides the empirical foundation for privacy claims.

Approach: Perform statistical characterisation of both anchor datasets including load profile visualisation, peak detection, and clustering by consumption pattern. Apply simple occupancy inference methods to both datasets to assess whether household level routines (sleeping time,

working hours, home/away patterns) are detectable in synthetic data. Tools: Python (pandas, matplotlib, scipy, scikit-learn).

Links to gaps: Provides empirical grounding for the attribute disclosure evaluation in Goal 2 and the GDPR assessment in Goal 3.

Goal 6: Recommendations and Evaluation Framework

Research question: What practical guidelines and evaluation standards should researchers using synthetic data in smart grid applications follow to ensure their work is ethically sound?

Motivation: The preceding goals produce empirical findings; this goal synthesises those findings into actionable recommendations. The literature currently lacks a community standard for the ethical evaluation and documentation of synthetic smart meter datasets, and the research is well positioned to contribute one based on empirical evidence.

Approach: Based on the findings of Goals 1–5, formulate a structured evaluation pipeline with defined stages (source data characterisation, statistical fidelity, NILM performance, privacy evaluation, bias audit, documentation). Develop a disclosure checklist for synthetic smart meter dataset publications. Validate the framework against the two anchor datasets.

Links to gaps: Synthesises findings from Gaps 1–6 into a practical contribution.

Conclusion

Before concluding, it is worth briefly reflecting on why i find this research direction genuinely interesting rather than merely assigned. Working through the literature for this review has sharpened my sense that synthetic data is neither the privacy panacea that some advocates claim nor the empty gesture that some critics suggest. It is a genuinely useful tool whose effectiveness depends critically on how it is applied and evaluated. The most important contribution this research can make is not to add another synthetic dataset to the literature but to contribute to the practices and standards by which synthetic datasets are evaluated, documented, and used responsibly. It feels satisfying to make these analysis and also learn deeply from my reviews and research project.

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